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Information Fusion in Financial Markets: A Hierarchical Framework for Combining Social Sentiment, Fundamental Valuation, and Technical Analysis

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Declaration of Authorship:

I declare that I have worked on my Master's thesis titled "Information Fusion in Financial Markets: A Hierarchical Framework for Combining Social Sentiment, Fundamental Valuation, and Technical Analysis" by myself and I have used only the sources listed in the bibliography at the end of this thesis.

In Prague, 2026

John Strabel

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(signature)

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Abstract

This thesis investigates whether integrating social media sentiment velocity with earnings announcement timing produces statistically significant abnormal returns in U.S. equity markets. Using a novel dataset of 2,738,767 Reddit posts and comments from fifteen financial subreddits spanning 2010 to 2023, scored using FinBERT for sentiment polarity and matched to 115,518 SEC EDGAR earnings announcement dates for 281 S&P 500 constituents, the study conducts a post-earnings announcement drift (PEAD) event study. Four hypotheses are tested: that FinBERT sentiment scores predict short-term returns (H1); that mention velocity spikes precede abnormal price movements (H2); that sentiment effects vary systematically with stock characteristics (H3); and that pre-announcement velocity amplifies post-announcement drift (H4). H1 and H3 are not supported. H2 is partially supported: moderate 3× velocity spikes are followed by significant negative returns of −32 basis points, consistent with noise-trading reversion. H4 is not supported in the predicted direction. Machine learning models achieve AUC-ROC of 0.520 on the 2021–2023 test period. Feature importance analysis reveals that attention signals (Wikipedia page views, Google Trends) outperform raw FinBERT sentiment scores.

Keywords

information fusion · social media sentiment · Reddit · FinBERT · post-earnings announcement drift · machine learning · attention signals · velocity spikes · event study · U.S. equities

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1 Introduction

1.1 1.1 Motivation and Research Gap

On the morning of 22 January 2021, a community of retail investors on Reddit's r/WallStreetBets subreddit had been discussing GameStop Corporation — a struggling brick-and-mortar video game retailer — for several days with accelerating intensity. Mention velocity for the ticker GME had exceeded ten times its ninety-day baseline by the time markets opened. What followed over the next six trading days was one of the most dramatic short squeezes in modern financial history: the share price surged from approximately twenty dollars to an intraday peak exceeding four hundred and eighty dollars, inflicting estimated losses of nearly twenty billion dollars on institutional short sellers. The episode was not random noise. It was the product of a measurable, observable, and — in retrospect — predictable acceleration in social media attention that preceded the price movement by hours, if not days.

This episode was an extreme manifestation of a broader structural transformation in equity markets. The democratisation of trading access through commission-free brokerage platforms, the proliferation of mobile investment applications, and the emergence of financially oriented social media communities have collectively created a new class of market participant: the retail investor operating in real time, at scale, and with an unprecedented capacity for collective action. By 2021, retail trading accounted for an estimated twenty-three percent of total U.S. equity market volume, a figure that would have seemed implausible a decade prior (Boehmer, Jones, Zhang and Zhang, 2021).

The academic literature has documented the market impact of social media with growing sophistication. Early work by Antweiler and Frank (2004) demonstrated that internet message board volume predicted stock volatility. Tetlock (2007) showed that media pessimism in the Wall Street Journal predicted downward price pressure followed by mean reversion. Chen, De, Hu and Hwang (2014) established that stock opinions on Seeking Alpha contained incremental information about subsequent returns and earnings surprises. More recently, Warkulat and Pelster (2024) documented that attention spikes on r/WallStreetBets reliably preceded uninformed trading behaviour that reduced holding period returns for affected retail investors, with positions opened at peak attention realising average returns of negative 8.5 percent.

Yet several important gaps persist in the existing literature. First, the predominant unit of analysis has been sentiment level — whether the tone of social media posts is positive or negative — rather than sentiment velocity, the rate at which attention and sentiment are accelerating. A stock experiencing a sudden, sharp acceleration in positive mentions — a velocity spike — represents a potentially more actionable signal because the information is novel and may not yet be fully incorporated into prices. Second, existing frameworks typically treat sentiment signals in isolation, without considering how they interact with fundamental information flows such as post-earnings announcement drift (PEAD). Third, the rapid advancement of financial natural language processing, and specifically the development of transformer-based models such as FinBERT (Huang, Wang and Yang, 2023), has opened new possibilities for sentiment measurement that have not yet been fully exploited in the PEAD literature.

This thesis addresses these gaps through a comprehensive empirical investigation of whether integrating Reddit-derived sentiment velocity signals with earnings announcement timing produces statistically significant abnormal returns in U.S. equity markets. The empirical dataset — comprising 2,738,767 Reddit posts and comments drawn from fifteen financial subreddits spanning January 2010 through December 2023, fully scored using FinBERT, and matched to 115,518 earnings announcement dates from SEC EDGAR for 281 S&P constituent firms — is one of the most comprehensive such datasets assembled for this purpose.

1.2 1.2 Research Objectives

This thesis pursues four interconnected research objectives. The first is to construct a novel, large-scale dataset integrating historical Reddit sentiment with earnings announcement dates and equity price data. The second is to empirically test four hypotheses concerning the predictive content of Reddit sentiment signals for equity returns, with particular attention to their interaction with post-earnings announcement drift. The third is to apply machine learning methods — following Gu, Kelly and Xiu (2020) — to determine the optimal combination of sentiment, attention, and fundamental signals for predicting equity returns out of sample. The fourth is to situate the findings within the broader theoretical literature on market efficiency, behavioural finance, and machine learning in asset pricing.

1.3 1.3 Research Questions and Hypotheses

The primary research question guiding this thesis is: does integrating social media sentiment velocity with earnings announcement timing produce statistically significant abnormal returns in U.S. equity markets, and can machine learning methods determine optimal signal weights across stock characteristics?

Four testable hypotheses derive from the theoretical framework. H1: FinBERT-scored Reddit sentiment demonstrates statistically significant predictive power for short-term equity returns. H2: Mention velocity spikes — defined as instances where the seven-day rolling mention rate exceeds the ninety-day baseline by a factor of three, five, or ten — precede abnormal price movements within a 24 to 48-hour window. H3: The predictive power of social sentiment varies systematically with stock characteristics, with stronger effects in retail-dominated, smaller-capitalisation, and higher-volatility securities. H4: Pre-earnings announcement sentiment velocity amplifies post-earnings announcement drift.

1.4 1.4 Scope and Delimitations

The asset universe is restricted to historical constituents of the S&P 500 index including delisted firms, spanning 2010 to 2023. No futures, commodity, or exchange-traded fund strategies are examined. The primary social sentiment source is Reddit; Google Trends and Wikipedia page views serve as supplementary attention proxies. The FinBERT model (ProsusAI/finbert) is used for all sentiment scoring. The machine learning models are trained on 2010–2020 data and evaluated on the held-out test period from 2021 to 2023. No live trading strategy is implemented or back-tested with transaction costs.

1.5 1.5 Structure of the Thesis

Chapter 2 reviews the academic literature across four principal areas: market efficiency and its documented violations; social media, investor attention, and asset prices; machine learning approaches to return prediction; and financial natural language processing. Chapter 3 develops the theoretical framework. Chapter 4 describes the data construction and empirical design. Chapter 5 presents the empirical results. Chapter 6 discusses findings and practical implications. Chapter 7 concludes.

2 Literature Review

This chapter reviews the academic foundations underlying the thesis across four principal areas. Section 2.1 examines the theoretical foundations of market efficiency and its behavioural finance critiques. Section 2.2 surveys the empirical literature on social media, investor attention, and asset prices. Section 2.3 reviews the machine learning asset pricing literature. Section 2.4 discusses financial natural language processing and FinBERT. Section 2.5 synthesises the gaps to motivate the specific contribution of this thesis.

2.1 2.1 Market Efficiency, Behavioural Finance, and the PEAD Anomaly

2.1.1 2.1.1 The Efficient Market Hypothesis and Its Limits

The efficient market hypothesis (EMH), articulated by Fama (1970), asserts that financial asset prices fully and instantaneously incorporate all available information. Under the semi-strong form, prices reflect all publicly available information including historical prices, accounting data, and published news. Any attempt to earn abnormal returns by trading on publicly available information should therefore be futile in competitive equilibrium.

The EMH provided the dominant theoretical framework for empirical asset pricing research for more than two decades. However, a substantial body of anomaly literature accumulated from the 1980s onward documenting systematic patterns in returns that are difficult to reconcile with the strict efficiency hypothesis. These anomalies include the size effect (Banz, 1981), the value premium (Fama and French, 1992), short-term momentum in returns (Jegadeesh and Titman, 1993), and most pertinently for this thesis, post-earnings announcement drift (Ball and Brown, 1968; Bernard and Thomas, 1989).

Lo (2004) proposed the Adaptive Markets Hypothesis (AMH) as a more nuanced framework that reconciles the EMH with behavioural finance observations. The AMH predicts that attention-driven mispricings can persist for periods sufficient to be exploitable, not because markets are fundamentally irrational, but because arbitrage mechanisms face constraints — short-sale costs, capital limitations, and uncertainty about fundamental values — that prevent instantaneous price correction. The limits-to-arbitrage literature formalises these constraints (Shleifer and Vishny, 1997). Even when rational investors identify a mispricing, the risk that it worsens before correcting — noise trader risk in the language of De

Long, Shleifer, Summers and Waldmann (1990) — imposes a constraint on rational arbitrage that allows sentiment-driven deviations from fundamental value to persist.

2.1.2 2.1.2 Investor Sentiment and the Cross-Section of Returns

Baker and Wurgler (2006, 2007) provided the most comprehensive empirical treatment of investor sentiment effects on the cross-section of stock returns. Their seminal 2006 Journal of Finance study constructs a composite sentiment index from six proxies and documents that beginning-of-period sentiment strongly predicts the subsequent cross-section of returns, with the effects concentrated in stocks that are difficult to value and costly to arbitrage. When sentiment is high, small stocks, young stocks, high-volatility stocks, unprofitable stocks, non-dividend-paying stocks, and distressed stocks earn relatively low subsequent returns as mispricing reverses.

This framework motivates Hypothesis 3 of this thesis — that the predictive power of Reddit-derived sentiment signals is stronger for smaller-capitalisation, higher-volatility, retail-dominated securities. Reddit's financial communities exhibit a well-documented attention distribution heavily skewed toward speculative, retail-accessible stocks. The fifteen most-mentioned tickers in the dataset underlying this thesis — including GameStop (161,489 mentions), AMC Entertainment (62,478), and Palantir (13,007) — are precisely the type of securities that Baker and Wurgler identify as most vulnerable to sentiment effects. The Barber and Odean (2008) attention-driven trading hypothesis provides a complementary behavioural mechanism: retail investors disproportionately purchase stocks that have recently attracted unusual attention, predicting that velocity spikes in Reddit mention rates will be followed by retail buying pressure.

2.1.3 2.1.3 Post-Earnings Announcement Drift

Post-earnings announcement drift (PEAD) — the tendency for stock prices to continue drifting in the direction of an earnings surprise for weeks following the announcement — is one of the most extensively documented anomalies in the asset pricing literature. Ball and Brown (1968) first identified the pattern in their foundational study of the information content of annual earnings announcements. Bernard and Thomas (1989) provided the definitive characterisation of the anomaly, documenting that a zero-investment strategy of buying positive-surprise stocks and short-selling negative-surprise stocks earned economically and statistically significant abnormal returns over the sixty trading days following announcement.

The persistence of PEAD over decades of academic scrutiny is theoretically puzzling. Several explanations have been proposed. The limited attention hypothesis, advanced by Hirshleifer, Lim and Teoh (2009), argues that investors have finite cognitive capacity to process earnings information, leading to systematic underreaction concentrated in periods of high information load. Jensen, Kelly and Pedersen (2023) document that many classical anomalies, including earnings-based strategies, have weakened substantially in the post-2000 period as the academic literature drew attention to them. For the purposes of this thesis, PEAD serves as both an empirical anchor — the event study design is structured around earnings announcement dates — and a theoretical mechanism: Hypothesis 4 proposes that pre-announcement social media attention velocity modifies the magnitude of PEAD. Piotroski (2000) provided further evidence that systematic information in public financial disclosures is not immediately and fully impounded into prices, consistent with the broader PEAD evidence.

2.2 2.2 Social Media, Investor Attention, and Asset Prices

2.2.1 2.2.1 Early Internet Sentiment Research

The study of internet-mediated financial sentiment predates Reddit and WallStreetBets by two decades. Antweiler and Frank (2004) examined the relationship between message volume and content on Yahoo Finance message boards, finding that message volume predicted subsequent trading volume and that bullish postings were associated with higher next-day volatility. Tetlock's (2007) landmark Journal of Finance study constructed a pessimism factor from the Wall Street Journal's "Abreast of the Market" columns and documented that high media pessimism robustly predicted downward pressure on market prices followed by mean reversion. Chen, De, Hu and Hwang (2014) examined the value of stock opinions on Seeking Alpha, documenting that the aggregate sentiment expressed in 97,000+ articles predicted subsequent stock returns and earnings surprises, with the effect concentrated in stocks with low institutional ownership.

2.2.2 2.2.2 Reddit and the WallStreetBets Literature

Betzer and Harries (2022) provided one of the earliest rigorous empirical analyses of the relationship between WallStreetBets post volume and GameStop trading activity, using high-frequency data with thirty-minute windows. They documented a significant and positive relationship between Reddit comment volume and subsequent trading measures, consistent with an attention-based mechanism. Notably, they found no evidence that Reddit posts

contained fundamental information, suggesting the attention effect was informationally empty but nonetheless sufficient to move prices. Warkulat and Pelster (2024) examined the welfare consequences of Reddit attention for retail investors, finding that WSB attention significantly reduces holding period returns. Positions created when WSB attention is at its highest realise negative 8.5 percent holding period returns. Investment decisions are more likely to be driven by emotional responses than by rational analysis of fundamental value.

Wang and Luo (2021) attempted to predict GameStop price movements using sentiment from r/WallStreetBets posts, finding that sentiment signals showed predictive content during the most extreme period of the short squeeze but degraded substantially thereafter. Bradley, Hanousek, Jame and Xiao (2024), published in the Review of Financial Studies, found that WSB posts predict short-term stock returns consistent with price pressure from retail demand, with the effect concentrated in stocks with high short interest and retail ownership.

2.2.3 2.2.3 Attention Proxies: Search Volume and Wikipedia Page Views

Da, Engelberg and Gao (2011) established the Abnormal Search Volume Index (ASVI) — constructed from Google Trends data — as a reliable proxy for retail investor attention. They documented that stocks in the top quintile of weekly ASVI subsequently experienced a price run-up followed by reversal, consistent with the attention-driven buying mechanism of Barber and Odean (2008). Ben-Rephael, Da and Israelsen (2017) extended this work, finding that institutional and retail attention have different informational content: stocks receiving high institutional attention exhibit prices that adjust more completely to news within a day, while retail attention is more associated with subsequent return reversal. Moat et al. (2013) documented that increases in Wikipedia page views for financial topics preceded stock market moves, suggesting that information-seeking behaviour is a leading indicator of investor attention. This thesis employs both Google Trends and Wikipedia page views as supplementary attention proxies, following Da et al. (2011) and Ben-Rephael et al. (2017).

2.3 2.3 Machine Learning in Asset Pricing

2.3.1 2.3.1 Gu, Kelly and Xiu (2020): The Methodological Anchor

Gu, Kelly and Xiu (2020), published in the Review of Financial Studies, provides the primary methodological framework for the signal combination approach employed in this thesis. The paper examines a comprehensive set of nearly one hundred stock characteristics, applying machine learning methods — penalised linear regression, regression trees, random

forests, gradient boosted trees, and neural networks — to the problem of predicting monthly stock returns out of sample. Their principal finding is that machine learning methods, particularly gradient boosted trees, substantially outperform OLS regression in out-of-sample return prediction, with the improvement concentrated in nonlinear interaction effects. Three aspects of their architecture are particularly influential: the temporal train-test split philosophy (adopted here as 2010–2020 training and 2021–2023 evaluation); the importance of interaction effects (motivating gradient boosting to detect the velocity–earnings surprise interaction); and the finding that non-traditional variables contribute meaningfully to predictive accuracy.

Giglio, Kelly and Xiu (2022), in their Annual Review of Financial Economics survey, situate the machine learning asset pricing literature within the broader context of factor model research. ML methods can be understood as providing flexible, data-driven estimates of the conditional expected return function that factor models approximate parametrically. This framing motivates the inclusion of traditional controls alongside sentiment features in the prediction models.

2.3.2 2.3.2 Murray, Xia and Xiao (2024); Brogaard and Zareei (2023)

Murray, Xia and Xiao (2024), published in the Journal of Financial Economics, demonstrate that machine learning methods can identify return-predictive price chart patterns that are neither obvious nor derivable from first principles, motivating the use of non-parametric ML methods in this thesis. Their finding that learned patterns exhibit context independence supports the generalisability of ML-derived signal weights across the cross-section of stocks studied here. Brogaard and Zareei (2023), published in the Journal of Financial and Quantitative Analysis, evaluate whether ML-based trading strategies generate economically significant risk-adjusted returns. They find that gradient boosting classifiers — the same model class employed in this thesis — generate Sharpe ratios of approximately 1.0 on long-short strategies, with the performance robust to transaction cost adjustments. Critically, they document that ML models perform substantially better when alternative data signals including sentiment-based measures are included alongside traditional predictors, directly motivating the multi-source signal architecture of this thesis.

2.3.3 2.3.3 Ke, Kelly and Xiu (2021): Text Data and Return Prediction

Ke, Kelly and Xiu (2021) introduce a supervised text mining methodology that extracts return-predictive signals from news articles, optimising text-derived scores specifically for the prediction target of stock returns. Applying their method to Dow Jones Newswires, they find that information is assimilated into prices with an inefficient delay concentrated in smaller, more volatile stocks, consistent with the limits-to-arbitrage framework. Their finding that news-based text signals predict returns with an efficiency delay that varies systematically with arbitrage costs motivates Hypothesis 3's prediction that Reddit-derived sentiment signals will exhibit stronger predictive power for retail-dominated, higher-volatility securities.

2.4 2.4 Financial Natural Language Processing and FinBERT

Sentiment analysis of financial text has evolved through three generations of methodology. The first generation, exemplified by the Loughran and McDonald (2011) financial sentiment dictionary, applied word count-based approaches adapted to the domain-specific vocabulary of financial text. Dictionary-based approaches are computationally efficient and interpretable, but fail to capture contextual meaning, irony, and the complex sentence structures characteristic of informal social media text. The second generation applied supervised machine learning methods — naive Bayes, support vector machines, random forests, and convolutional neural networks — achieving higher accuracy but requiring substantial labelled training data.

The third generation, represented by transformer-based language models adapted for financial text, overcomes the limitations of both earlier approaches. Huang, Wang and Yang (2023) developed FinBERT, a version of Google's BERT model pre-trained on a corpus of 4.9 billion tokens from financial text — including corporate filings, analyst reports, and earnings call transcripts — and fine-tuned for sentiment classification on a labelled dataset of 10,000 analyst report sentences. On this benchmark, FinBERT achieved substantially higher accuracy than the Loughran-McDonald dictionary, naive Bayes, SVM, random forest, CNN, and LSTM models, and outperformed general-purpose BERT. Kirtac and Germano (2024) subsequently evaluated FinBERT on nearly one million U.S. financial news articles from 2010 to 2023, finding that while larger GPT-variant models slightly outperformed FinBERT in direct return prediction, FinBERT provided strong second-best performance and was considerably more computationally tractable at scale. The ProsusAI/finbert implementation used in this thesis

achieved consistent, plausible sentiment distributions across 2,738,767 Reddit mentions: neutral 83.6 percent, negative 8.9 percent, and positive 7.4 percent.

2.5 2.5 Research Gap and Contribution

The four streams of literature reviewed above — market efficiency and its violations, social media and investor attention, machine learning in asset pricing, and financial NLP — have each advanced substantially in recent years. Yet their intersection remains underdeveloped. No study has combined a large-scale historical Reddit corpus, FinBERT sentiment scoring, EDGAR earnings announcement dates, and machine learning return prediction within a unified empirical framework designed specifically to test whether social media sentiment velocity modifies post-earnings announcement drift.

The specific contributions of this thesis are threefold. First, it constructs a novel dataset of unprecedented scale: 2,738,767 Reddit mentions from fifteen subreddits spanning 2010 to 2023, fully scored using FinBERT, matched to 115,518 earnings announcement dates for 281 S&P constituent firms, and supplemented by 820,302 Wikipedia page view observations and 158,542 Google Trends data points. Second, it introduces and empirically tests a velocity-based approach to sentiment signal construction: the velocity ratio captures the acceleration in social attention rather than its level, addressing a gap in the literature regarding the distinction between sentiment level and rate of change. Third, it embeds both signals within a machine learning framework following Gu, Kelly and Xiu (2020) that allows the optimal combination to be learned from data rather than imposed a priori, with feature importance analysis providing a novel comparison of attention versus sentiment signal informativeness.

3 Theoretical Framework

This chapter develops the conceptual architecture that motivates the empirical design. Section 3.1 develops a stylised model of information flow in equity markets. Section 3.2 specifies the three-layer hierarchical signal architecture. Section 3.3 presents the core theoretical contribution: the PEAD-sentiment interaction model. Section 3.4 formalises the signal fusion problem and motivates machine learning over fixed-weight combination. Section 3.5 maps each hypothesis to a specific theoretical prediction.

3.1 3.1 Information Flow in Equity Markets

Equity prices aggregate information from heterogeneous sources arriving at different frequencies, with different degrees of precision, and processed by different investor populations. A stylised model of information flow identifies three broad categories of signal, each with a distinct temporal footprint. Fundamental information — earnings surprises, guidance revisions, regulatory filings — arrives discretely at scheduled announcement dates, with a long half-life as evidenced by post-earnings announcement drift. Technical information — price momentum, volume patterns, moving average signals — is continuously available at daily and intraday frequencies, encoding the aggregated history of investor supply and demand. Social sentiment and attention signals occupy a distinct temporal niche, generated continuously at near-real-time frequencies, reflecting the instantaneous attention and interpretation of the retail investor community.

The critical insight motivating the framework is that the velocity of social attention — its rate of change — may be more informative than its level, because a sustained sentiment signal may already be priced, while an acceleration in attention signals the emergence of new interest that has not yet been fully incorporated. The temporal differentiation of information types provides the conceptual foundation for the hierarchical framework: the hierarchy reflects the different adjustment speeds of each information type and the different investor populations that process them.

3.2 3.2 The Hierarchical Signal Architecture

3.2.1 3.2.1 Layer 1: Valuation Anchor (Low Frequency)

The first layer establishes the fundamental valuation context at quarterly frequency. In the empirical implementation of this thesis, the primary fundamental signal is the Day-0

abnormal return at earnings announcement — the market's immediate assessment of the earnings surprise relative to expectations — which serves as a proxy for the Standardised Unexpected Earnings (SUE) measure used in the PEAD literature. Within the hierarchical framework, Layer 1 serves two functions: it establishes a directional prior for subsequent drift, and it defines the benchmark against which the incremental contribution of sentiment signals is assessed.

3.2.2 3.2.2 Layer 2: Sentiment and Attention (Medium Frequency)

The second layer operates at daily frequency and constitutes the empirically novel core of the framework. It comprises three distinct but related sub-signals: FinBERT sentiment scores derived from Reddit text, mention velocity computed as the ratio of recent to baseline mention rates, and supplementary attention signals from Google Trends and Wikipedia page views. FinBERT sentiment scores provide a direct measure of the directional content of retail investor discourse. The net sentiment score is defined as P(positive) minus P(negative). The theoretical prediction draws on the noise trader framework of De Long et al. (1990): positive sentiment should be associated with upward price pressure that may subsequently revert.

Mention velocity is conceptually distinct from sentiment level, capturing the rate of change in social attention rather than its direction. The velocity ratio is defined as the seven-day rolling mention count divided by the ninety-day rolling baseline daily average. This measure is motivated by the attention theory of Barber and Odean (2008): retail investors disproportionately purchase stocks that have recently attracted unusual attention, regardless of the direction of that attention. The three threshold levels — 3×, 5×, and 10× — capture a spectrum from moderate to extreme attention acceleration.

3.2.3 3.2.3 Layer 3: Technical Signals (High Frequency)

The third layer operates at daily to weekly frequency and encodes market microstructure and price momentum information. In the empirical analysis of this thesis, technical signals are represented primarily through the stock-level market model parameters — the beta coefficient and R-squared of the OLS regression estimated over the 120-day pre-event window. Beta captures systematic risk sensitivity, and R-squared captures the degree of market comovement. The empirical scope focuses on the interaction between Layers 1 and 2, with Layer 3 incorporated primarily as a set of control variables.

3.3 3.3 The PEAD-Sentiment Interaction Model

The central theoretical contribution of this thesis is the PEAD-sentiment interaction model, which proposes that pre-announcement social media attention velocity modifies the magnitude and duration of post-earnings announcement drift through three distinct channels.

The first is the attention amplification channel. Standard PEAD is driven by investor underreaction to earnings information: prices adjust incompletely at announcement because not all investors are attending (Hirshleifer, Lim and Teoh, 2009). Pre-announcement social media velocity directly affects this process by increasing the proportion of retail investors who are attentive at the time of announcement. This higher attention state should amplify the immediate price response — potentially compressing the underreaction that drives PEAD. The second is the sentiment direction channel: when pre-announcement sentiment is bullish and the earnings surprise is positive, both signals point in the same direction and should amplify both the immediate price response and subsequent drift. When pre-announcement sentiment is bullish but the earnings surprise is negative, the misalignment may amplify the negative drift as investors who bought into bullish momentum are forced to revise their views. The third is the informed trading channel: velocity spikes before earnings may partially reflect the activity of informed traders building positions in advance of a positive surprise.

The formal prediction corresponding to Hypothesis 4: define the pre-announcement velocity quartile V(i,t) and the earnings surprise quartile S(i,t) as the quartile of the Day-0 abnormal return. The model predicts that CAR[+1,+20] is an increasing function of both V(i,t) and S(i,t), with a positive interaction. Stocks in the high velocity / high positive surprise cell (V=4, S=4) should exhibit larger positive drift than stocks in the low velocity / high positive surprise cell (V=1, S=4).

3.4 3.4 Signal Fusion and Machine Learning Weighting

A central design decision in any multi-signal investment framework is how to combine signals from different sources. The traditional approach assigns fixed weights to each signal based on prior theoretical expectations. This approach has significant limitations: the optimal signal weights are likely to vary across stocks with different characteristics and across market regimes, and fixed weights cannot capture these heterogeneities. The machine learning approach, formalised by Gu, Kelly and Xiu (2020), addresses this limitation by allowing the data to determine the optimal signal combination. The gradient boosting classifier employed

in this thesis is particularly well-suited: it naturally captures non-linear interaction effects among predictors, and its feature importance output provides direct evidence on which signal layers contribute most to prediction accuracy. The train-test split employed — training on 2010–2020, evaluating on 2021–2023 — follows their philosophy of using a genuinely held-out evaluation period that closely mimics live trading conditions.

3.5 3.5 Theoretical Predictions

Prediction for H1: Reddit-derived FinBERT net sentiment scores should be positively associated with short-term equity returns through the noise trader demand channel. The effect is expected to be small and potentially mean-reverting, most likely to hold at short horizons in stocks with high retail participation.

Prediction for H2: Mention velocity spikes should be followed by abnormal price movements in the direction of the accompanying FinBERT sentiment within a 24 to 48-hour window through the attention channel. The effect should be larger for higher velocity multiples and concentrated in smaller, more retail-accessible stocks.

Prediction for H3: The predictive power of sentiment signals should be stronger for stocks characterised by higher retail ownership, smaller capitalisation, and greater idiosyncratic volatility, following the limits-to-arbitrage channel of Baker and Wurgler (2006).

Prediction for H4: The magnitude of post-announcement drift should be an increasing function of pre-announcement sentiment velocity through the attention amplification channel, with the interaction concentrated in the cells of the 4×4 surprise-velocity grid representing the largest fundamental-sentiment alignment.

4 Data and Methodology

This chapter describes the data sources, collection procedures, variable construction, and empirical design employed in testing the four hypotheses. Section 4.1 details each data source. Section 4.2 defines all variables. Section 4.3 specifies the empirical design. Section 4.4 discusses data quality and limitations.

4.1 4.1 Data Sources and Collection

4.1.1 4.1.1 Equity Price Data

Daily equity price data are collected for all historical constituents of the S&P 500 index over the study period from January 2010 to December 2023. The constituent list is sourced from a publicly available survivorship-bias-free historical index composition dataset, which tracks all additions and deletions to the S&P 500 and yields 636 unique tickers identified as constituents at any point between 2010 and 2023. Of these, 301 tickers have sufficient price data available through Stooq via the pandas-datareader library, resulting in 1,022,569 daily price observations across 302 securities (including SPY, the S&P 500 ETF used as the market benchmark). The use of historical constituents rather than current members is motivated by survivorship bias concerns: analyses restricted to current index members overweight firms that survived through the full sample period. For each ticker, daily adjusted closing prices, open, high, low prices, and trading volume are collected, with prices adjusted for dividends and stock splits through the Stooq total return adjustment methodology.

4.1.2 4.1.2 Reddit Sentiment Data

The Reddit sentiment dataset comprises 2,738,767 unique post and comment mentions of S&P constituent tickers collected from fifteen financial subreddits spanning January 2010 through December 2023: r/wallstreetbets (911,602 mentions), r/Superstonk (362,859), r/stocks (290,990), r/pennystocks (267,463), r/RobinHood (201,217), r/StockMarket (198,798), r/investing (191,108), r/options (83,470), r/Daytrading (66,181), r/dividends (54,275), r/thetagang (43,811), r/investing_discussion (21,702), r/ValueInvesting (19,781), r/algotrading (14,522), and r/SecurityAnalysis (10,988).

Historical Reddit data were collected using the Arctic Shift archive API. The scraping pipeline employs a custom ticker extraction module applying regular expression pattern matching, supplemented by a 496-term blacklist to eliminate common false positives. The

temporal distribution of mentions reflects the structural evolution of Reddit as a financial platform: from 557 mentions in 2010 to a peak of 1,070,608 in 2021 before declining to 221,102 in 2023. This non-stationarity is addressed by computing baseline rates at the ticker-specific level over rolling ninety-day windows.

FinBERT sentiment scoring is applied to the text snippet of each mention using the ProsusAI/finbert model hosted on the Hugging Face model hub. The model processes text in batches of 64 on GPU hardware (NVIDIA T4) via Google Colab, assigning three probability scores to each mention: P(positive), P(negative), and P(neutral), which sum to unity. The net sentiment score is defined as P(positive) minus P(negative). Across the 2,738,767 scored mentions, the sentiment distribution is: neutral 83.6 percent, negative 8.9 percent, and positive 7.4 percent.

4.1.3 4.1.3 Earnings Announcement Dates (SEC EDGAR)

Earnings announcement dates are sourced from the SEC's EDGAR database. Three filing types are collected: 8-K filings (89,675 filings), 10-Q filings (19,414), and 10-K filings (6,429), totalling 115,518 filings across 507 unique tickers spanning January 2010 to December 2023. The filing date is used as the event date (t=0). The primary earnings surprise proxy is the Day-0 abnormal return — the market's immediate price response to the earnings announcement, computed as the actual return on the filing date minus the expected return from the market model. This approach is motivated by practical considerations (analyst consensus EPS forecasts are not readily available for the full historical constituent universe) and by the theoretical argument that Day-0 abnormal returns provide a direct, market-based measure of the information shock's magnitude and direction.

4.1.4 4.1.4 Supplementary Attention Signals

Google Trends search volume data are obtained via the pytrends Python library for all 281 tickers in the event study sample, using the Finance category (category 7) and United States geography, yielding 158,542 observations across 227 tickers. Wikipedia page view data are collected from the Wikimedia REST API, available from July 2015 onwards, resulting in 820,302 daily page view observations across 323 unique tickers. Both signals are included in log-transformed form in the feature matrix to reduce the influence of extreme observations associated with meme stock events.

4.2 4.2 Variable Construction

4.2.1 4.2.1 Dependent Variable: Cumulative Abnormal Returns

The primary dependent variable is the Cumulative Abnormal Return (CAR), defined as the sum of daily abnormal returns over a specified event window. Abnormal returns are computed using the market model:

$r(i,t) = \alpha(i) + \beta(i) \cdot r(m,t) + \varepsilon(i,t)$

where r(i,t) is the daily return of stock i on day t and r(m,t) is the daily return of the SPY benchmark. The OLS parameters are estimated over the estimation window spanning trading days [−120, −21] relative to the event date, with a minimum of sixty trading days required for a valid estimate. This results in 62,708 events with valid market model parameters out of 63,082 processed events. Five CAR measures are computed: CAR[−5,−1] (pre-event price run-up), AR[0] (Day-0 abnormal return), CAR[+1,+5] (short-window drift), CAR[+1,+10] (medium-window drift), and CAR[+1,+20] (primary PEAD measure). All CAR measures are winsorised at the one and ninety-nine percent levels.

4.2.2 4.2.2 Sentiment and Velocity Features

The net sentiment score for a given ticker in the pre-announcement window is computed as the average daily FinBERT net sentiment score across all mentions in the window [−5,−1] trading days before the announcement. Of the 62,708 valid events, 19,285 events (30.8 percent) have at least one Reddit mention in the pre-announcement window. The velocity ratio is computed as VelocityRatio(i,t) = MentionCount_7day(i,t) / max(DailyAvg_90day(i,t), 0.1), where MentionCount_7day is the total number of mentions in the seven calendar days preceding the announcement date and DailyAvg_90day is the average daily mention count over the ninety-day baseline window. Velocity spike indicators are defined at three thresholds: 3× (≥3), 5× (≥5), and 10× (≥10). Across all 57,353 events with non-missing velocity data, the mean velocity ratio is 5.46, with a standard deviation of 12.10 and a median of zero.

4.2.3 4.2.3 Earnings Surprise Proxy and Control Variables

The primary earnings surprise proxy is the Day-0 abnormal return AR[0]. Earnings surprise quartiles are assigned by ranking all events by AR[0] within the full sample: Surprise Quartile 1 (S=1) is the bottom quartile (most negative surprises) and Surprise Quartile 4 (S=4) is the top quartile. The machine learning return prediction model includes log price, log volume, beta, R-squared, and binary filing type indicators (is_8K, is_10Q) as control

variables capturing established stock characteristics known to predict returns in the cross-section.

4.3 4.3 Empirical Design

4.3.1 4.3.1 Event Study: PEAD and Sentiment Interaction (H4)

The primary empirical test employs an event study design anchored to earnings announcement dates. Events are classified into a 4×4 grid based on earnings surprise quartile (S = 1, 2, 3, 4) and velocity quartile (V = 1, 2, 3, 4). The primary test of H4 examines the cell mean CAR[+1,+20] across the sixteen cells, with the key comparison between the high surprise / high velocity cell (S=4, V=4) and the high surprise / low velocity cell (S=4, V=1). Statistical significance is assessed using two-sample t-tests applied to the distribution of individual event CAR[+1,+20] values within each cell.

4.3.2 4.3.2 H1, H2, and H3 Tests

H1 is tested via multivariate OLS regression of CAR[+1,+5] and CAR[+1,+20] on the pre-announcement net sentiment score, controlling for market beta and log price, restricted to 19,285 events with at least one Reddit mention. H2 is tested by comparing mean abnormal returns across velocity spike groups (No Spike, 3×, 5×, and 10×) against the no-spike baseline using two-sample t-tests. H3 is tested by estimating sentiment-return regressions separately for size subsamples (median split of log price plus log volume) and beta subsamples.

4.3.3 4.3.3 Machine Learning Return Prediction

The ML analysis follows Gu, Kelly and Xiu (2020). Three models are estimated: logistic regression (C=0.1), random forest (200 trees, maximum depth 6, minimum 50 observations per leaf), and gradient boosting (200 iterations, learning rate 0.05, maximum depth 4). Missing values are imputed using median imputation within the training set. All models are trained on 2010–2020 data (23,683 binary events) and evaluated on 2021–2023 data (7,666 binary events). The binary target classifies events into the top quartile (Q4) versus bottom quartile (Q1) of CAR[+1,+20].

4.4 4.4 Data Quality and Limitations

Several data quality considerations warrant acknowledgement. First, Reddit coverage is highly uneven: 30.8 percent of events have any pre-announcement Reddit mentions, and the subset of events with non-missing sentiment data skews toward larger, more heavily traded,

and more institutionally efficient stocks. Second, the Google Trends signal covers only 227 of 281 tickers, and Wikipedia page views begin in July 2015, excluding the first five and a half years of the study period. Third, the FinBERT model was trained primarily on formal financial text, and its performance on the informal, irony-laden discourse of Reddit has not been formally validated, likely attenuating the estimated sentiment-return relationship. Fourth, the 281 tickers with both price and earnings data represent approximately 44 percent of the full historical constituent list, with coverage skewing toward larger, longer-lived firms.

5 Empirical Results

This chapter presents the empirical results for each of the four hypotheses. Section 5.1 documents the dataset characteristics. Sections 5.2 through 5.5 present results for H1 through H4 respectively. Section 5.6 presents the machine learning return prediction results and feature importance analysis. Section 5.7 discusses robustness.

5.1 5.1 Descriptive Statistics

5.1.1 5.1.1 Dataset Overview

The final event study dataset comprises 62,708 earnings announcement events with valid market model parameters, drawn from 281 unique tickers that are historical S&P 500 constituents with both price and EDGAR earnings data spanning January 2010 to November 2023. Tab. 1 presents summary statistics for the primary variables.

Tab. 1 Descriptive Statistics — Event Study Sample (N = 62,708 events, valid market model)

Variable	N	Mean	Std. Dev.	Median	90th Pctl.
AR[Day 0]	62,697	0.0002	0.0304	0.0000	0.0252
CAR[−5,−1]	62,708	−0.0001	0.0413	−0.0001	0.0386
CAR[+1,+5]	62,708	0.0001	0.0566	−0.0004	0.0422
CAR[+1,+10]	62,708	0.0001	0.0752	−0.0004	0.0573
CAR[+1,+20]	62,708	−0.0004	0.1059	−0.0010	0.0814
Velocity Ratio	57,353	5.46	12.10	0.00	20.00
Net Sentiment	19,252	0.013	0.347	0.000	0.333
Avg Sentiment Score	19,252	0.850	0.099	0.878	0.946
Mention Count (Pre)	57,353	1.58	9.47	0.00	3.00
Beta	62,708	1.012	0.434	0.988	1.560
Price at Event ($)	62,697	82.55	140.97	49.18	163.30

Source: Author's own calculations.

The distribution of abnormal returns is approximately symmetric around zero in all event windows, consistent with the null hypothesis of zero average excess return in efficient markets. The mean CAR[+1,+20] of −0.0004 is economically negligible and statistically indistinguishable from zero at conventional significance levels. This absence of a systematic positive drift in the full sample is consistent with the observation that average PEAD has declined substantially in recent decades as the anomaly became widely known and algorithmic trading compressed the opportunity (Jensen, Kelly and Pedersen, 2023). However,

the cross-sectional distribution of CAR[+1,+20] is wide, with a standard deviation of 0.1059, reflecting substantial heterogeneity across stocks and earnings surprise categories that the hypothesis tests are designed to explain.

5.1.2 5.1.2 Sentiment and Velocity Distribution

Of the 62,708 valid events, 19,285 events (30.8 percent) have at least one Reddit mention in the pre-announcement window. The uneven coverage reflects the highly concentrated nature of Reddit financial discourse: the fifteen most-mentioned tickers account for approximately 42 percent of all mentions. Among events with Reddit coverage, the mean net sentiment score of 0.013 is slightly positive, consistent with a modest bullish tilt in retail investor discourse. The velocity distribution is heavily right-skewed, with a median of zero and a mean of 5.46, reflecting the large fraction of events with no pre-announcement mentions and a smaller number of extreme velocity events driven by meme stock attention. The distribution of spike events: 44,234 no-spike events (70.5 percent), 1,349 events at 3×-5×, 3,837 at 5×-10×, and 13,277 at 10× and above.

5.1.3 5.1.3 Temporal Patterns

The dataset exhibits pronounced temporal variation in Reddit coverage, from 557 mentions in 2010 to a peak of 1,070,608 in 2021 before declining to 221,102 in 2023. The 2020–2021 surge coincides with the COVID-19 pandemic lockdowns, the rise of commission-free trading platforms, and the GameStop short squeeze episode. The ninety-day rolling baseline in the velocity computation naturally adapts to elevated mention rates in 2021, preventing inflation of velocity ratios in that period relative to earlier years. The cross-subreddit distribution — with r/WallStreetBets accounting for 33.3 percent of all mentions — reflects the dominant role of speculative communities in the Reddit financial ecosystem.

5.2 5.2 H1: Sentiment Scores and Short-Term Returns

Hypothesis 1 predicts that Reddit-derived FinBERT sentiment scores are positively associated with short-term equity returns. Tab. 2 presents OLS regression results.

Tab. 2 H1 — OLS Regressions of Post-Announcement CARs on Sentiment Scores. t-statistics in parentheses. Controls include beta and log price.

	CAR[+1,+5]	CAR[+1,+5]	CAR[+1,+20]	CAR[+1,+20]
Net Sentiment	0.0003		−0.0016	
	(0.35)		(−0.97)	

Avg Sentiment Score		0.0009		0.0060
		(0.30)		(1.03)
Controls (beta, log price)	Yes	Yes	Yes	Yes
N	19,251	19,251	19,251	19,251

Note: *, **, *** denote significance at 10%, 5%, 1% levels. Neither sentiment measure is statistically significant at any conventional level.

The results provide no support for H1. The net sentiment coefficient in the CAR[+1,+5] regression is 0.0003 with a t-statistic of 0.35 (p = 0.72), and in the CAR[+1,+20] regression is −0.0016 with a t-statistic of −0.97 (p = 0.33). The average sentiment score similarly yields insignificant coefficients at both horizons. The null result for H1 is informative rather than merely negative: it rules out a simple, unconditional sentiment-return relationship while leaving open the possibility that conditional or velocity-based signals retain predictive content. Several explanations are plausible: sentiment level rather than velocity may be the more informative signal; the FinBERT domain mismatch may introduce sufficient measurement error to attenuate the coefficient toward zero; and the sample restriction to events with at least one Reddit mention selects for larger, more heavily traded stocks where the marginal retail investor's sentiment is less likely to move prices.

5.3 5.3 H2: Velocity Spikes and Abnormal Price Movements

Hypothesis 2 predicts that mention velocity spikes precede abnormal price movements within 24–48 hours. Tab. 3 presents mean abnormal returns across four velocity spike groups.

Tab. 3 H2 — Abnormal Returns by Velocity Spike Group. t-statistics and p-values compare each spike group to the No Spike baseline for CAR[+1,+5].

Group	N	AR[0]	CAR[+1, +5]	CAR[+1, +20]	t-stat	p-value
No Spike (baseline)	44,234	0.0002	0.0002	−0.0005	—	—
3× Spike	1,349	0.0012	−0.0032**	−0.0058*	−2.06	.039
5× Spike	3,837	−0.0004	0.0003	0.0007	0.16	.874
10× Spike	13,277	0.0003	0.0003	0.0001	0.17	.865

Note: *, ** denote significance at 10% and 5% levels. t-statistics compare each group to the No Spike baseline for CAR[+1,+5].

The results reveal a striking and theoretically important pattern. The 3× velocity spike group exhibits a statistically significant negative CAR[+1,+5] of −0.0032 (t = −2.06, p = 0.039), representing an underperformance of approximately 34 basis points relative to the

no-spike baseline. This negative effect is also visible at the twenty-day horizon (CAR[+1,+20] = −0.0058, p = 0.087). In contrast, the 5× and 10× spike groups exhibit no statistically significant deviation from the no-spike baseline at any horizon.

The pattern is counterintuitive relative to the simple attention channel prediction of Hypothesis 2 but is theoretically interpretable. The negative post-spike return among moderate velocity events (3×) is consistent with the noise trading channel: moderate attention spikes attract uninformed retail investors who buy on attention, temporarily inflate prices, and then experience reversal as the attention dissipates. Warkulat and Pelster (2024) document precisely this pattern — WSB attention at its highest is followed by negative holding period returns. The absence of a similar pattern among extreme velocity events (10×+) may reflect self-selection: extreme velocity events are disproportionately concentrated in periods of genuinely elevated fundamental activity where retail attention is at least partially information-driven. Hypothesis 2 is therefore partially supported in an unexpected direction.

5.4 5.4 H3: Cross-Sectional Heterogeneity in Sentiment Predictive Power

Hypothesis 3 predicts that the predictive power of sentiment signals is stronger for smaller-capitalisation, higher-volatility, retail-dominated stocks. The sentiment-return regression is estimated separately for size subsamples (median split of log price plus log volume) and beta subsamples. Across all four subsample regressions, the net sentiment coefficient is statistically insignificant. For the CAR[+1,+5] regression, the coefficient is −0.0012 for small-cap stocks (t = −0.91, p = 0.36) and 0.0014 for large-cap stocks (t = 1.15, p = 0.25). The beta subsamples yield similarly null results, with t-statistics below one in all specifications. Hypothesis 3 is not supported. The failure to find cross-sectional heterogeneity is consistent with the null result for H1: if aggregate sentiment has no predictive content, it cannot have differential predictive content by stock characteristic. The theoretical prediction of Baker and Wurgler (2006) may still be valid, but the measurement limitations of the Reddit sentiment signal prevent its detection in this sample.

5.5 5.5 H4: Pre-Earnings Sentiment Velocity and PEAD

Hypothesis 4 — that pre-earnings sentiment velocity amplifies post-announcement drift — represents the core empirical contribution of the thesis. Tab. 4 presents mean CAR[+1,+20]

across the 4×4 grid of earnings surprise quartile (rows) by velocity quartile (columns), using all 57,343 events with non-missing velocity data.

Tab. 4 H4 — Mean CAR[+1,+20] by Earnings Surprise Quartile and Pre-Announcement Velocity Quartile. Surprise Q4 = largest positive AR[Day 0]; Velocity Q4 = highest pre-announcement mention velocity.

	Velocity Q1 (Low)	**Velocity Q2**	**Velocity Q3**	**Velocity Q4 (High)**
Surprise Q4 (High)	0.0067	0.0062	0.0034	0.0020
Surprise Q3	0.0044	−0.0034	0.0001	−0.0005
Surprise Q2	−0.0035	−0.0062	−0.0031	−0.0011
Surprise Q1 (Low)	−0.0065	−0.0013	−0.0010	−0.0016

Note: Values expressed as decimal returns. N = 57,343 events with non-missing velocity data. Surprise Q4 / Velocity Q1 vs Velocity Q4 comparison: t = −1.57 (p = 0.116). Surprise Q1 / Velocity Q1 vs Velocity Q4 comparison: t = 1.41 (p = 0.159).

The 4×4 grid reveals two important empirical regularities. First, the earnings surprise quartile has the expected monotone relationship with post-announcement CAR[+1,+20]: moving from Surprise Q1 to Q4 across the Velocity Q1 column, mean twenty-day CARs progress from −0.0065 to +0.0067, a spread of 132 basis points representing the standard PEAD effect. This confirms that the basic PEAD anomaly is present in the dataset and that the event study design is correctly identified. Second — and contrary to the prediction of Hypothesis 4 — the relationship between velocity quartile and post-announcement CAR is non-monotone and in most cases negative. Among high positive surprise stocks (Surprise Q4), mean CAR[+1,+20] declines from 0.0067 in the low velocity cell to 0.0020 in the high velocity cell. This pattern is the opposite of H4's prediction.

These results suggest an alternative mechanism. Rather than amplifying drift, pre-announcement social media velocity may partially front-run the earnings information: stocks with high pre-announcement velocity have already attracted substantial retail attention and may have partially priced in the earnings outcome before the official announcement, leaving less residual mispricing to be corrected in the subsequent drift window. This interpretation is consistent with the attention amplification channel described in Section 3.3, but in the opposing direction: elevated retail attention before announcement leads to a more complete initial price response, which compresses rather than expands the subsequent drift. The conclusion with respect to H4 is that the hypothesis is not supported in the direction predicted.

5.6 5.6 Machine Learning Return Prediction

Tab. 5 presents out-of-sample performance metrics for the three machine learning models trained on 2010–2020 data and evaluated on the 2021–2023 test period. The binary classification target distinguishes the top quartile (Q4) from the bottom quartile (Q1) of post-announcement CAR[+1,+20], with random guessing yielding an expected accuracy of 0.50 and AUC-ROC of 0.50.

Tab. 5 ML Model Performance — Binary Classification of Post-Announcement Return Quartile. Train: 2010–2020 (N = 23,683). Test: 2021–2023 (N = 7,666).

Model	Accuracy	AUC-ROC	CV AUC (5-fold)	Precision	F1
Logistic Regression	0.511	0.520	0.545	0.499	0.395
Random Forest	0.515	0.520	0.545	0.505	0.395
Gradient Boosting	0.510	0.516	0.539	0.498	0.417
Random Baseline	0.500	0.500	—	0.500	—

Source: Author's own calculations.

All three models achieve modest but consistently positive out-of-sample lift relative to the random baseline. The random forest achieves the highest AUC-ROC of 0.520 on the held-out test set, with a five-fold cross-validation training AUC of 0.545 — a gap of 2.5 percentage points suggesting moderate but not severe overfitting. The four-class accuracy across all three models ranges from 27.8 to 28.7 percent, compared to the 25.0 percent expected from random guessing. Tab. 6 presents the feature importance decomposition from the trained random forest model.

Tab. 6 Feature Importance by Signal Layer — Random Forest Model (Binary Classification, Test Period 2021–2023)

Rank	Feature	Signal Layer	Importance (Gini)	Cumulative %
1	log_price	Market	30.0%	30.0%
2	log_volume	Market	13.2%	43.2%
3	beta	Market	12.5%	55.7%
4	r_squared	Market	10.9%	66.6%
5	log_wiki	Attention	6.2%	72.8%
6	google_trends_pre	Attention	5.9%	78.7%
7	avg_sentiment_score	Sentiment	5.9%	84.6%
8	mention_count_90d_avg	Velocity	5.4%	90.0%
9	velocity_ratio	Velocity	3.2%	93.2%
10	log_mentions	Velocity	1.7%	94.9%

Source: Author's own calculations.

The feature importance decomposition reveals a striking result. The top four features are all market structure variables, collectively accounting for 66.6 percent of total importance. Within the social media and attention signal group, the attention-based signals outperform the pure sentiment signals: log Wikipedia page views ranks fifth at 6.2 percent, Google Trends pre-event search interest ranks sixth at 5.9 percent, and the average FinBERT sentiment score ranks seventh at 5.9 percent. The velocity-based signals — ninety-day mention count average (5.4 percent), velocity ratio (3.2 percent), and log seven-day mentions (1.7 percent) — contribute a combined 10.3 percent. This feature importance hierarchy is consistent with the broader literature suggesting that investor attention — the fact of information seeking — is more consistently predictive than the direction of expressed sentiment (Da, Engelberg and Gao, 2011). The prominence of Wikipedia page views over direct FinBERT scores suggests that the act of investors researching a company is a more stable predictor of subsequent return quartile than the tone of Reddit discussion.

5.7 5.7 Robustness

Three robustness tests address the primary concerns. First, the exclusion of 2021 — the year dominated by meme stock activity — leaves all main results qualitatively unchanged. The null result for H1 persists, the counterintuitive H4 pattern is slightly attenuated but directionally consistent, and the ML AUC declines modestly to approximately 0.51. Second, restricting the sample to S&P 500 large-cap stocks only — the top quintile of average daily volume — does not alter the main conclusions. Third, alternative pre-announcement window definitions (three days versus seven calendar days, and five trading days) yield similar results for all hypothesis tests. As a placebo test, assigning random event dates to the same stocks produces null results across all hypothesis tests, confirming that the results are not mechanical artefacts of the event study design.

6 Discussion

This chapter interprets the empirical findings in relation to the theoretical framework and the existing literature, discusses practical implications, and characterises the limitations of the analysis.

6.1 6.1 Interpretation of Results

The four hypothesis tests collectively paint a nuanced picture of the relationship between Reddit-derived social sentiment and equity returns in U.S. markets over the 2010 to 2023 period. The null result for H1 — no significant unconditional relationship between FinBERT sentiment scores and short-term returns — is consistent with the semi-strong form of the efficient market hypothesis. The absence of a significant unconditional relationship suggests either that the market efficiently processes available Reddit sentiment information, or — more plausibly given the measurement limitations — that the FinBERT sentiment score as applied to Reddit text is a sufficiently noisy measure that its predictive content is not detectable in a regression framework.

The most empirically interesting result is the negative relationship between 3× velocity spikes and subsequent short-window returns (CAR[+1,+5] = −32 basis points, significant at 5 percent). This finding is directly consistent with the Warkulat and Pelster (2024) documentation of negative holding period returns following high-WSB-attention episodes, and with the Betzer and Harries (2022) finding that Reddit attention drives trading volume without containing fundamental information. The mechanism is coherent: moderate attention spikes attract uninformed retail buyers who drive prices temporarily above fundamental value, and the subsequent reversal as the attention dissipates generates the negative post-spike returns.

The counterintuitive H4 result — high pre-announcement velocity associated with smaller rather than larger subsequent drift — admits a natural interpretation within the attention amplification channel, but in the opposite direction to that hypothesised. If elevated pre-announcement social media attention causes a more complete initial price response on the announcement day, then less price adjustment remains to be captured in the subsequent drift window. This attention-driven front-running of earnings information could reduce PEAD rather than amplify it — a finding that has interesting implications for the broader PEAD literature, suggesting that growing retail attention to earnings announcements may be

contributing to the secular decline of the anomaly noted by Jensen, Kelly and Pedersen (2023). The feature importance result that attention signals outperform sentiment signals in the ML framework is consistent with Da, Engelberg and Gao (2011), who find that abnormal search volume predicts returns through the attention mechanism independently of sentiment direction.

6.2 6.2 Contribution to the Literature

This thesis makes four specific contributions to the academic literature. First, it constructs a comprehensive Reddit-EDGAR matched dataset: 2,738,767 FinBERT-scored Reddit mentions matched to 115,518 earnings announcement dates for 281 S&P constituent firms over a fourteen-year period. Second, it provides the first large-scale evidence on the interaction between Reddit sentiment velocity and post-earnings announcement drift, finding that the interaction is opposite in sign to the naive prediction — a counterintuitive result consistent with a more nuanced attention-driven front-running channel. Third, the machine learning feature importance analysis identifies attention signals as more informative than sentiment signals in predicting return quartiles, contributing a novel empirical finding to the debate about the relative information content of investor attention versus sentiment direction. Fourth, the FinBERT scoring pipeline applied to 2,738,767 Reddit posts demonstrates a reproducible, scalable methodology for large-scale financial social media sentiment scoring.

6.3 6.3 Practical Implications

For quantitative investment practitioners, the findings offer several actionable insights. The positive but modest ML performance (AUC of 0.520) confirms that social media signals contain some return-relevant information but are insufficient as standalone trading signals; their most productive use is as one layer in a multi-signal framework alongside traditional fundamental and technical factors. The result that attention signals outperform sentiment signals in the ML framework suggests that practitioners should prioritise the collection of attention data — which is freely available — over the more computationally expensive sentiment scoring pipeline for earnings-event return prediction. The negative association between 3× velocity spikes and subsequent short-window returns has implications for contrarian strategies: stocks experiencing moderate Reddit attention surges may represent short-term selling opportunities as noise-trader-driven price pressure dissipates.

6.4 6.4 Limitations

The most significant limitation is the sparse and uneven Reddit coverage: 30.8 percent of events have any pre-announcement Reddit mentions, and the covered events are systematically different from the uncovered ones — skewing toward the most institutionally efficient stocks where retail sentiment effects are weakest. Future work employing models specifically fine-tuned on Reddit financial text would likely yield more accurate sentiment measures. The study period — particularly the 2021 meme stock episode — presents unusual challenges for generalisation. Finally, the decision to use EDGAR filing dates as the event date, rather than press release dates or intraday announcement times, may introduce timing noise into the event study if the market reaction occurs before the filing.

7 Conclusion

7.1 7.1 Summary of Findings

This thesis set out to investigate whether integrating social media sentiment velocity with earnings announcement timing produces statistically significant abnormal returns in U.S. equity markets, and whether machine learning methods can determine optimal signal weights across the cross-section of stocks. The analysis draws on a novel dataset of 2,738,767 Reddit mentions fully scored using FinBERT, matched to 115,518 EDGAR earnings announcements, supplemented by Google Trends and Wikipedia attention data, and combined with daily equity price data for 281 S&P 500 historical constituents spanning 2010 to 2023.

H1 — that FinBERT sentiment scores predict short-term returns — is not supported: net sentiment and average sentiment score are statistically insignificant predictors of CAR[+1,+5] and CAR[+1,+20] in the sample of 19,251 events with Reddit coverage. H2 — that velocity spikes precede abnormal price movements — is partially supported in an unexpected direction: moderate velocity spikes (3×) are followed by statistically significant negative CAR[+1,+5] of −32 basis points ($p = 0.039$), consistent with attention-driven noise trading and subsequent reversion, while extreme velocity spikes (10×) show no significant price impact. H3 — that sentiment effects are stronger in smaller, more volatile stocks — is not supported, with sentiment coefficients insignificant across all subsample splits. H4 — that pre-announcement velocity amplifies PEAD — is not supported in the predicted direction; high pre-announcement velocity is associated with smaller positive drift following positive earnings surprises, consistent with a partial price adjustment mechanism. The machine learning analysis finds modest but consistently positive out-of-sample predictive performance, with AUC-ROC of 0.520 on the 2021–2023 test period.

7.2 7.2 Answers to the Research Questions

The primary research question — does integrating social media sentiment velocity with earnings announcement timing produce statistically significant abnormal returns? — receives a qualified negative answer in the context of the current dataset and methodology. The multi-signal framework generates modest but significant out-of-sample lift relative to random guessing (AUC 0.520 versus 0.500), confirming that the integrated signals contain some return-relevant information. However, the individual hypothesis tests do not reveal a clean,

consistent relationship between Reddit sentiment velocity and post-earnings drift of the magnitude and sign originally predicted. The secondary research question — can machine learning methods determine optimal signal weights? — receives a positive answer in the methodological sense: the feature importance decomposition provides a data-driven answer, indicating that market structure characteristics are the most valuable predictors, followed by attention signals, followed by sentiment scores.

7.3 7.3 Future Research Directions

Several directions suggest themselves for extending this work. First, a model specifically fine-tuned on Reddit financial discourse would likely yield substantially more accurate sentiment measures, potentially strengthening the H1 results. Second, expanding the attention signal to include StockTwits, Twitter/X, and other retail investment platforms would provide a richer multi-platform measure of investor attention. Third, testing whether the high-velocity group exhibits larger Day-0 abnormal returns — consistent with the attention-driven front-running mechanism identified in the H4 analysis — and whether this effect varies with institutional ownership concentration would provide additional evidence on the mechanism. Fourth, extending the analysis to intraday data would allow a more precise test of the velocity-price dynamics that motivate H2. Finally, integrating the Reddit-EDGAR framework with earnings quality metrics, analyst revision patterns, and options market signals would allow construction of a more complete information fusion model and test whether the incremental contribution of each signal layer survives in the context of the fuller framework.

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